CASE 07 · AI-AUGMENTED BUSINESS ANALYST PROJECT · HR & TALENT ACQUISITION

The 65% Blind Spot in AI Fraud Screening

Validating a zero-shot LLM job-posting fraud screen against confirmed EMSCAD fraud outcomes — a pilot report for a Talent Acquisition analytics team.

Role simulated: HR Analytics / Talent Acquisition Business Analyst | Tools: Python, LLM-based zero-shot classification (manual/agentic review), scikit-learn-style evaluation metrics

Source: EMSCAD Employment Scam Aegean Dataset (public, 17,880 real job postings, 2012–2014) — stratified evaluation sample of n=100 (40 confirmed-fraud + 60 confirmed-legitimate)

74%	35%	69%	48%
Precision — when the screen flags a posting, how often it's right	Recall — share of confirmed fraud the screen actually caught	Overall agreement with confirmed ground truth	F1 score — balance of precision and recall

The business problem

Fraudulent job postings damage two things at once: applicants who lose time, personal data, or money to scam recruiters, and the employer brand of any careers site or job board that unknowingly hosts them. A Talent Acquisition analytics team wants to know whether an LLM can serve as a first-pass fraud screen — reading a posting's text the way a recruiter would, and flagging suspicious ones for priority human review before they go live. This report is that pilot's validation study: does the screen's judgment agree with confirmed fraud outcomes well enough to justify deploying it as a triage aid?

Data recency note, stated plainly: the EMSCAD postings were scraped in 2012–2014. Specific scam phrasing may have shifted since (crypto and remote-work lures, for instance, weren't yet common), so absolute language patterns should be revalidated on current postings before production use — but the analytical method, and the core finding that text-only screening has real limits, both transfer.

The evaluation

A stratified sample of 100 postings was drawn with a fixed random seed: 40 confirmed-fraudulent and 60 confirmed-legitimate, deliberately oversampling the fraud class so the evaluation set is informative rather than dominated by negatives. This is stated plainly because it is not the population rate: fraud makes up only 4.8% of the full 17,880-posting dataset. For each of the 100 postings, a text blob (title + company profile + description + requirements + benefits, truncated to ~1,200 characters) was built and read blind — the true fraud label was withheld from the judging step. Each posting was judged individually, the way a recruiter using an AI copilot would: fraud suspected (yes/no), plus one to three specific red-flag reasons grounded in that posting's actual wording where a flag was raised. Those 100 independent judgments were then scored

against the real EMSCAD labels.

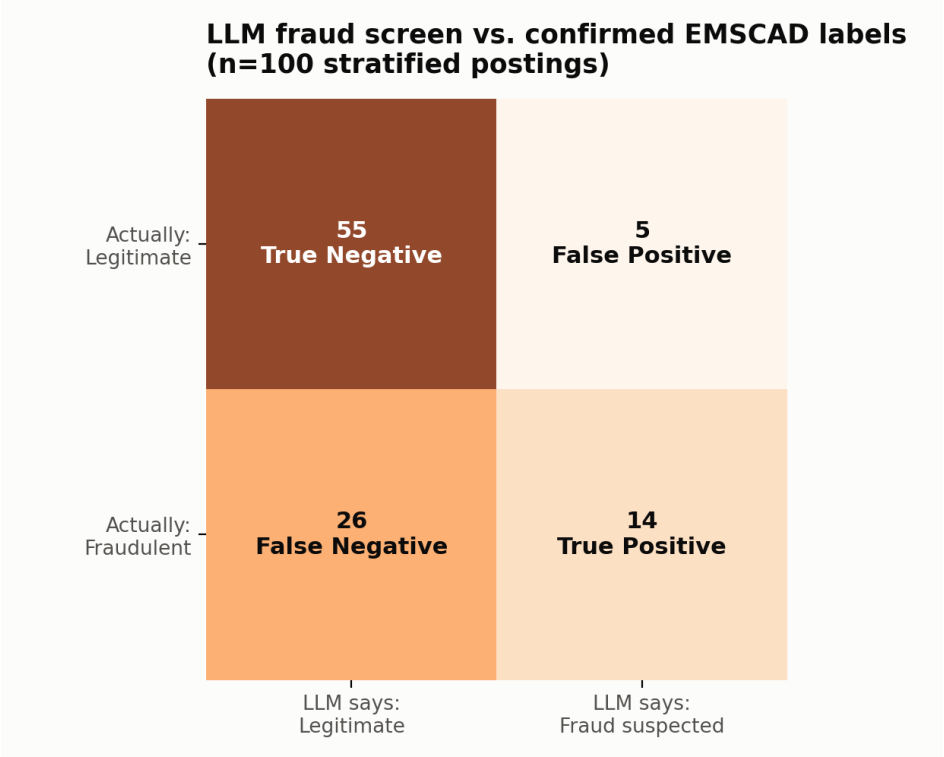


Fig. 1 — The screen's verdicts against confirmed EMSCAD labels on all 100 evaluated postings. It caught 14 of 40 confirmed-fraudulent postings and correctly cleared 55 of 60 legitimate ones.

What the LLM actually flags

The screen flagged 19 of the 100 postings. Its stated reasons cluster around a small set of recurring themes, tallied here from the screen's own written red-flag reasons rather than a predefined keyword list.

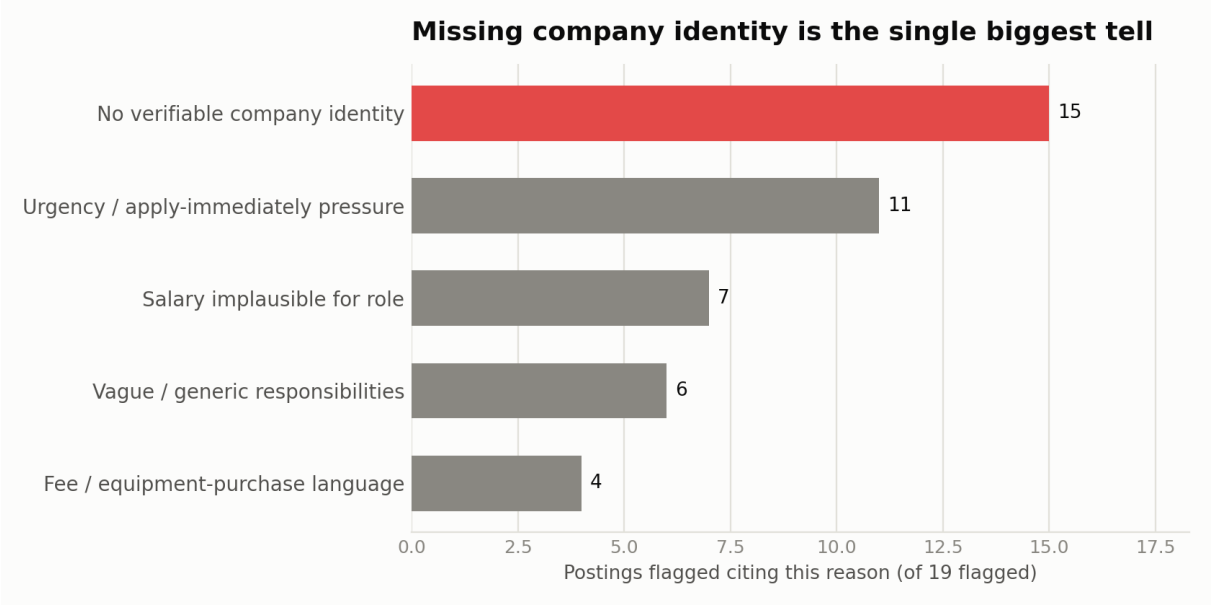


Fig. 2 — Red-flag themes across all flagged postings, grouped from the screen's own written rationales.

A few real examples, quoted directly from the screen's own judgments: for *“Home Based Payroll Typist/Data Entry Clerks Positions Available,”* it wrote — “$1,000-$2,500/week for unskilled home typing with 'no experience needed', plus 'GET STARTED RIGHT NOW' pressure and only a bare email as contact.” For *“customer service rep,”* a two-line posting, it wrote — “no company, 'needed asap', and 'will explain on phone interview' — deliberately withholds detail to move contact off-platform.” And for *“Network Marketing,”* it flagged the “implausible '$600-115,000 a month' range” and “recruit-and-earn framing.” In each case the reasoning is specific to that posting's actual text, not a generic template.

Where it gets it wrong

Accuracy of 69% sounds respectable, but the two error types tell very different stories, and both argue for keeping a human in the loop.

False negatives (26 of 40 confirmed frauds missed, a 65% miss rate) are the more important failure mode. Several missed postings referenced real, checkable organizations — a named hotel brand, a publicly traded telecom, a regional IT consultancy, a real utility company — with professional, technically coherent job descriptions. The screen's judgment leaned heavily on “does this read like a real company wrote it,” and sophisticated fraud that copies or impersonates a real employer's voice defeats that heuristic almost by design. Text alone cannot verify that a posting genuinely came from the company it names.

False positives (5 legitimate postings wrongly flagged) tell the opposite story. Four of the five were the same “English Teacher Abroad” posting template — informal tone, a first-name-only recruiter contact, generic benefit bullets — which matched the screen's mental model of a scam almost perfectly, but was in fact a real (if casually written) overseas recruiting agency. Surface-level informality and a real employer's actual marketing voice can look identical from text alone.

Together these errors are the clearest evidence in this report that a zero-shot text screen should not auto-reject or auto-approve anything. It's a strong prioritization signal with a meaningfully high false-negative rate on well-disguised fraud — exactly the profile that argues for human-in-the-loop triage, not full automation.

Recommendation

Deploy the LLM screen as a **priority-review triage layer**, not a gatekeeper. Concretely: every new posting is scored before publication; postings flagged 'fraud suspected' are routed to a human reviewer's priority queue with the screen's specific red-flag reasons attached (not just a score), so the reviewer can verify or dismiss quickly; postings the screen clears still publish on the normal timeline, since at 35% recall a clear result is not strong enough evidence of legitimacy to skip review entirely for high-risk categories (remote-only roles, unusually high pay, no named employer). The value case is queue triage, not decision automation: at 74% precision, a reviewer acting on a flag is right roughly three times out of four — a real efficiency gain over reviewing every posting unprioritized — while the human step catches the brand-impersonation fraud that text alone misses.

RESUME BULLET, READY TO USE

“Designed and ran a zero-shot LLM fraud-screening pilot against 100 real job postings from the EMSCAD dataset, scoring the screen at 74% precision / 35% recall against confirmed fraud outcomes; identified brand-impersonation false negatives as the key production risk and recommended a human-in-the-loop triage deployment over full automation.”

What's in this package

File	What it is
run_analysis.py	Scores the LLM judgments against ground truth and builds both charts — fully reproducible from llm_judgments.json.
llm_judgments.json	All 100 per-posting verdicts, red-flag reasons, and rationales, written individually against the text alone.
This report (PDF)	Methodology, confusion matrix, red-flag themes, error analysis, and the triage recommendation.